Aperture Science Portal Physics: Advanced Technical Documentation

Introduction

This document covers advanced physics concepts behind portal technology, intended for senior researchers and facilities engineers.

Quantum Tunneling Theory

Basic Principle

Portals operate by creating a traversable wormhole between two points in spacetime, requiring:

Exotic matter concentration (Moon-dust based catalyst)

Energy threshold of 1.5 MeV per portal

Perfect surface alignment (0.01 degree tolerance)

Portal Types

1. **Blue Portal**: Entrance portal (Creates stable wormhole entrance)
2. **Orange Portal**: Exit portal (Creates stable wormhole exit)
3. **Quantum Portal**: Dual-ended wormhole (Requires extreme conditions)

Physics Equations

Portal Momentum Conservation

Equation: $p_{total} = p_{initial}$ (Momentum preserved through portals)

Implications: Fall speed maintains through transit

Energy Requirements

Formula: $E = m \times c^2 \times (1/2 (v^2/c^2))$

For Portal Gun: Requires 1.2×10^{15} watts (Brief exposure)

Surface Requirements

Porosity: >95% moon dust content

Temperature: Room temperature (20 degrees C optimal)

Electrical Resistance: <0.01 ohms

Advanced Applications

Momentum Transfer

Objects retain velocity through portal

Enables high-speed travel across distances

Utilized for: Rapid delivery, particle acceleration

Quantum Entanglement

Portals are linked through quantum states

Information travels instantaneously between portals
Phone call possible through portal (If stable enough)

Bending Light

Light refracts through portal edges
Creates unique visual distortions
Used for: Optical illusions, emergency signaling

Portal Interaction Mechanics

Surface Types

Surface Type	Portal Capability
Moon-dusted white	Full capability
Standard wall	Limited (Temporary)
Glass	Unsuitable
Moving Platforms	Acceptable (Requires recalibration)

Portal Stability Factors

1. Distance between portals
2. Atmospheric pressure differentials
3. Quantum interference patterns
4. User mood (Handshake protocol)

Experimental Applications

Dual-Exit Portals

Two exit portals, one entrance
Requires advanced quantum splitting
Currently in development (Risk of fracturing)

Time-Dilated Portals

Research in temporal manipulation
Side effects include: Paradox formation
Not approved for civilian use

Safety Warnings

DO NOT create portal to same surface
Avoid creation near volatile materials
Report any anomalies to GLaDOS
Moon-dust surfaces (Must remain clean)